

Quick Start Guide

for
WAVE Device

Revision: 1.1

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1. Objective

This document provides a quick glance at Unex WAVE OBU device. It also demonstrates quick tests on sending and receiving packets, checking MIB status, upgrading firmware, and etc.

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2. Scope

Supported product:

» OBU-201U

» RSU-101U

For RSU at a glance, please refer to “RSU-101 Hardware Guide”.

Although targeting to OBU, the quick test demonstrations and CLI commands are suitable for both OBU and RSU due to sharing the same SDK.

Supported protocol stack:

» Unex WAVE-SDK-v1007 or above (apply Autotalk SDK-4.11.1)

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3. Abbreviations

OBU	Onboard Unit
RSU	Roadside Unit
DSRC	Dedicated Short Range Communication
CCH	Control Channel
SCH	Service Channel
PSID	Provider Service Identifier
WAVE	Wireless Access in Vehicular Environments
WSA	WAVE Service Advertisements
WME	WAVE Management Entity
BSM	Basic Safety Message
SPAT	Signal Phase And Timing Message
MAP	Map Data
TIM	Traveler Information Message
RSA	Road Side Alert
RTCM	Radio Technical Commission for Maritime Services
CSR	Common Safety Request Message
NMEA	National Marine Electronics Association
ICA	Intersection Collision Alert Message
EVA	Emergency Vehicle Alert Message
PDM	Probe Data Message
SSM	Signal Status Message
SRM	Signal Request Message

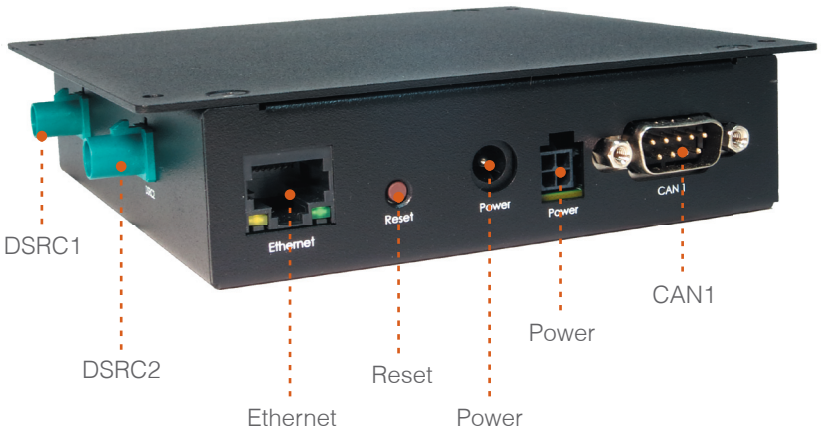
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PVD	Probe Vehicle Data Message
PSM	Pedestrian Safety Message

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4. OBU at a Glance

Front View:



Rear View:



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Accessories

Following accessories are included with OBU-201:

1. Round jack power adapter:



2. 4-pin automotive power connector:



3. GPS antenna:



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4. DSRC antennas:



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5. Get Started – Environment Setup

Prerequisites:

1. A PC with
   a. USB port
   b. USB to UART cable (ex. Prolific PL2030) with driver installed on PC
   c. Terminal emulator application (ex. PuTTY) installed on PC, baud rate set to 115200 bps.
2. Connect the key components (DSRC antennas, GPS antenna, USB to UART cable, and power adapter) as Figure 1



Figure 1 - Environment Setup

3. For firmware upgrade, an additional ethernet cable need to be connected between PC and WAVE device. Or connect both ethernet ports to a common switch/hub. The IP set in PC and WAVE device should be in the same sub-net. For example, 192.168.1.x, for both.

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Note:

For application flexibility in lab and vehicle, two kinds of power cables are provided. One is the round jack power adapter and the other one is 4-pin automotive power connector. It is recommended to use any ONE of the power cables at one time.

After device booted and terminal opened successfully on PC, a prompt "`ate>`" will be displayed in terminal. All CLI commands should be issued under "`ate`" mode

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6. WSM - Tx/Rx Demonstration

This demonstration will create channel service on channel 172, then transmit/receive WSM on this channel. It requires configuring two OBU-201 units, and only radio number 1 is used. Figure 2 illustrates the testing overview. For detail CLI command description, refer to “Application Programming Interface Guide for WAVE Device”.

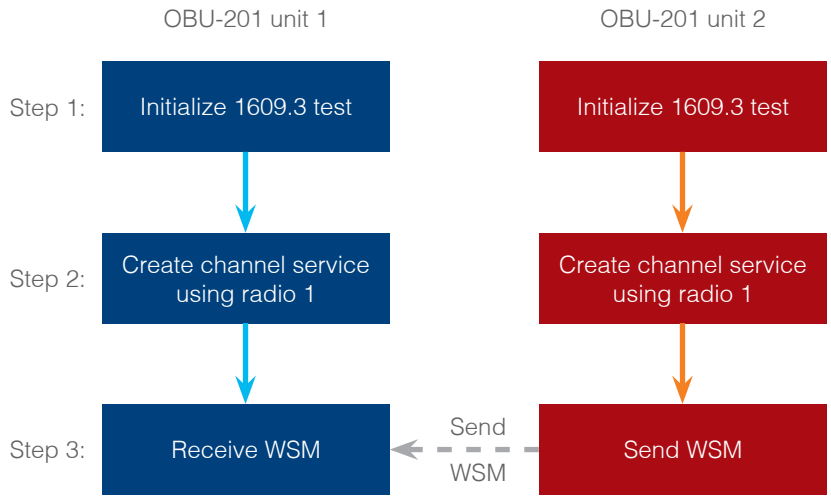


Figure 2 - Channel & WSM services Overview

Step 1:

On OBU-201 unit 1, issue following command to initialize 1609.3 test environment

```
dot3_test_init
```

This command returns a handle id from 1609.3 protocol stack to register other services such as: Provider, User, Channel, WSM

Figure 3 displays the snapshot

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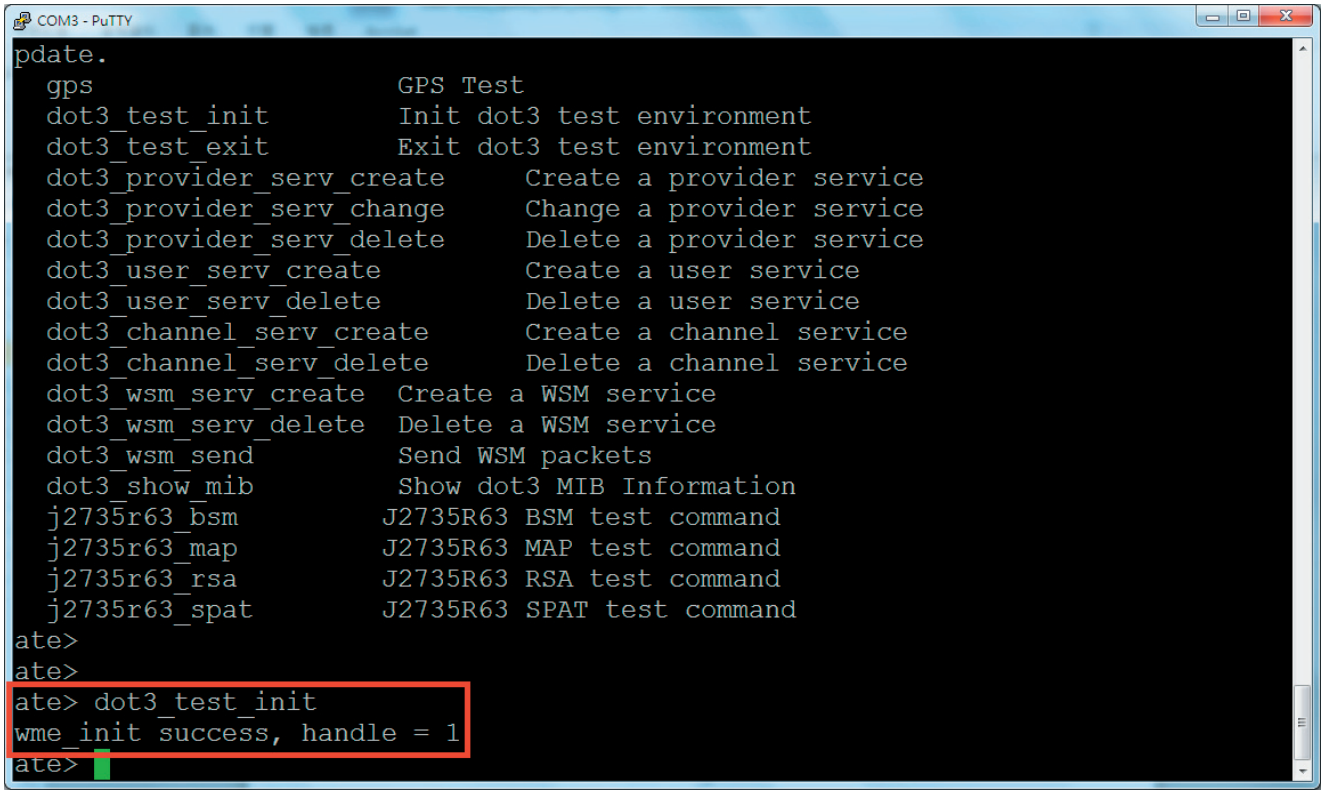


Figure 3 - dot3_test_init Snapshot

On OBU-201 unit 2, apply same command to initialize 1609.3 test environment

Step 2:

On OBU-201 unit 1, issue following command to create a channel service

```
dot3_channel_serv_create 1 1 172 3 5
```

This command applies handle 1 to create a channel service on Channel 172 using radio number 1. On success, a service index will be returned.

Figure 4 displays the snapshot

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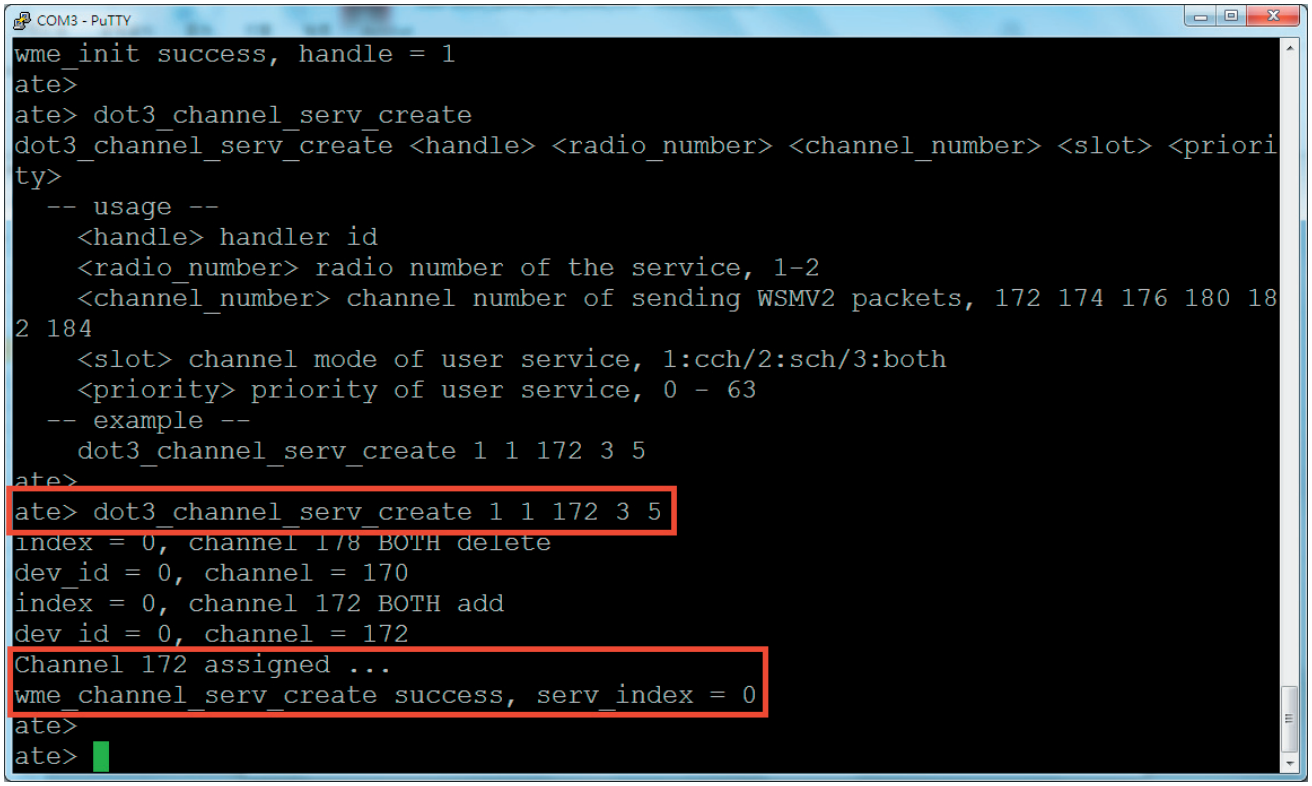


Figure 4 - dot3_channel_serv_create Snapshot

On OBU-201 unit 2, apply same command to create a channel service

Note:

After testing, we can issue following command (apply handle and service index) to delete channel service on both device.

```
dot3_channel_serv_delete 1 0
```

Step 3:

On OBU-201 unit 1, issue following command create a WSM service

```
dot3_wsm_serv_create 1 123
```

This command will receive all WSM packets with PSID 123

Figure 5 displays the snapshot

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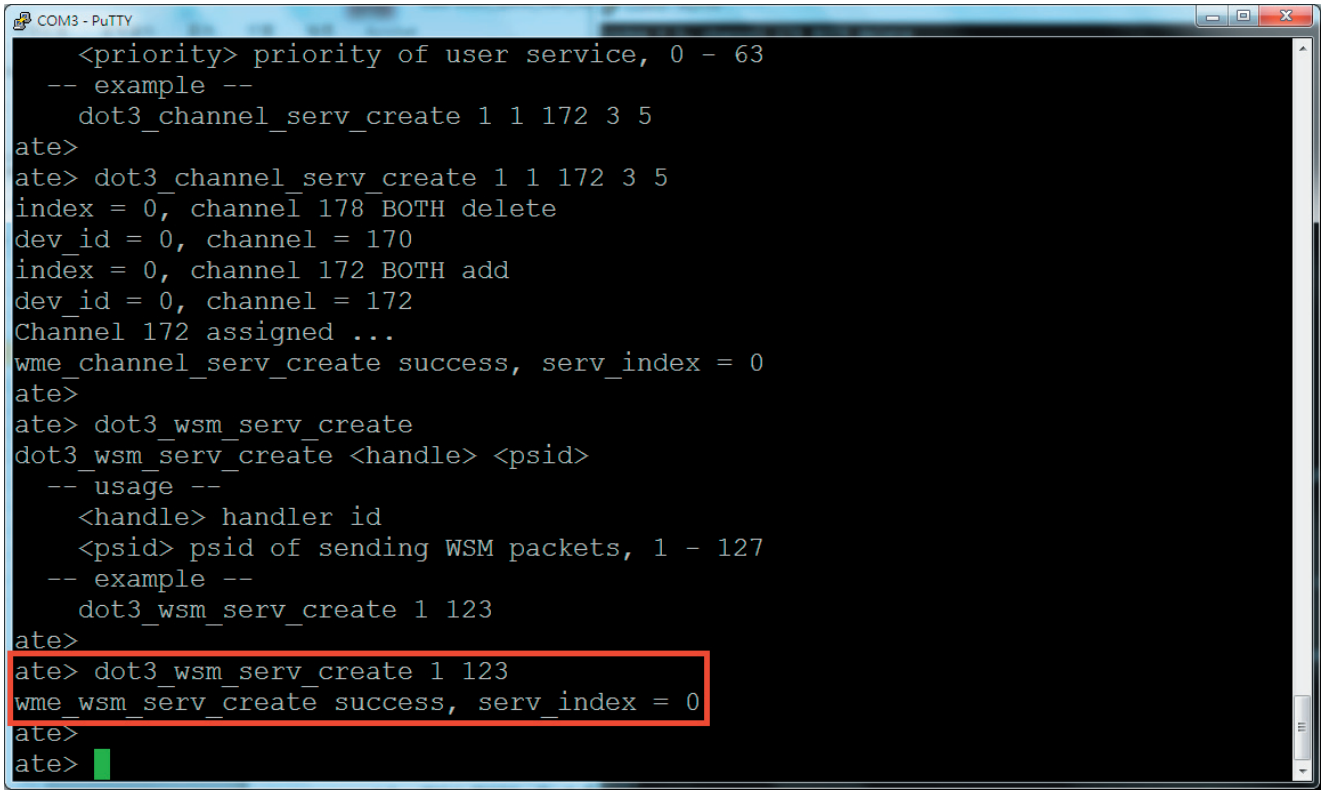


Figure 5 dot3_wsm_serv_create Snapshot

Note:

After testing, we can issue following command (apply handle and service index) to delete WSM service on both device.

```
dot3_wsm_serv_delete 1 0
```

On OBU-201 unit 2, issue following command to send WSM packets using radio number 1

```
dot3_wsm_send 1 123 1 172 6 18 FF:FF:FF:FF:FF:FF 1000 100 10
```

This command applies handle 1 to sends 10 WSM packets with PSID 123 on Channel 172 using radio number 1

Figure 6 displays the snapshot

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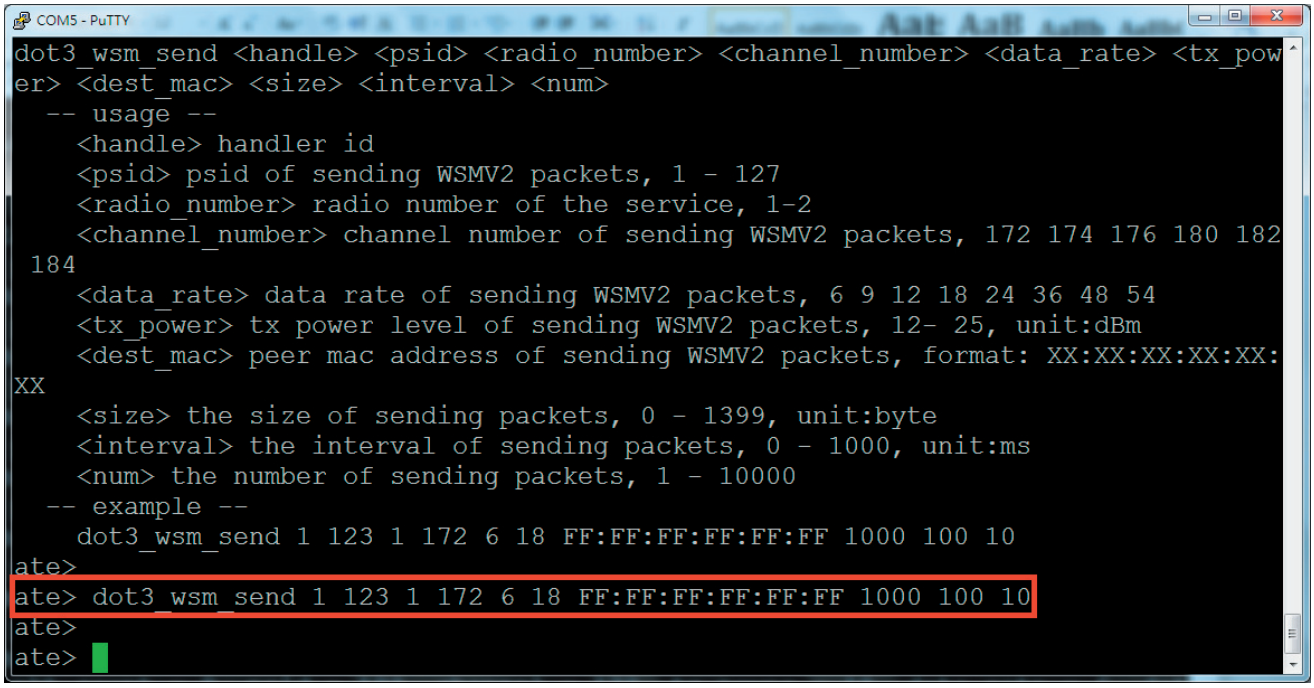


Figure 6 - dot3_wsm_send Snapshot

After execution of above steps, 10 WSM packets are successfully received on OBU-201 unit 1, as shown in Figure 7 `(recv pkt count = 10)`:

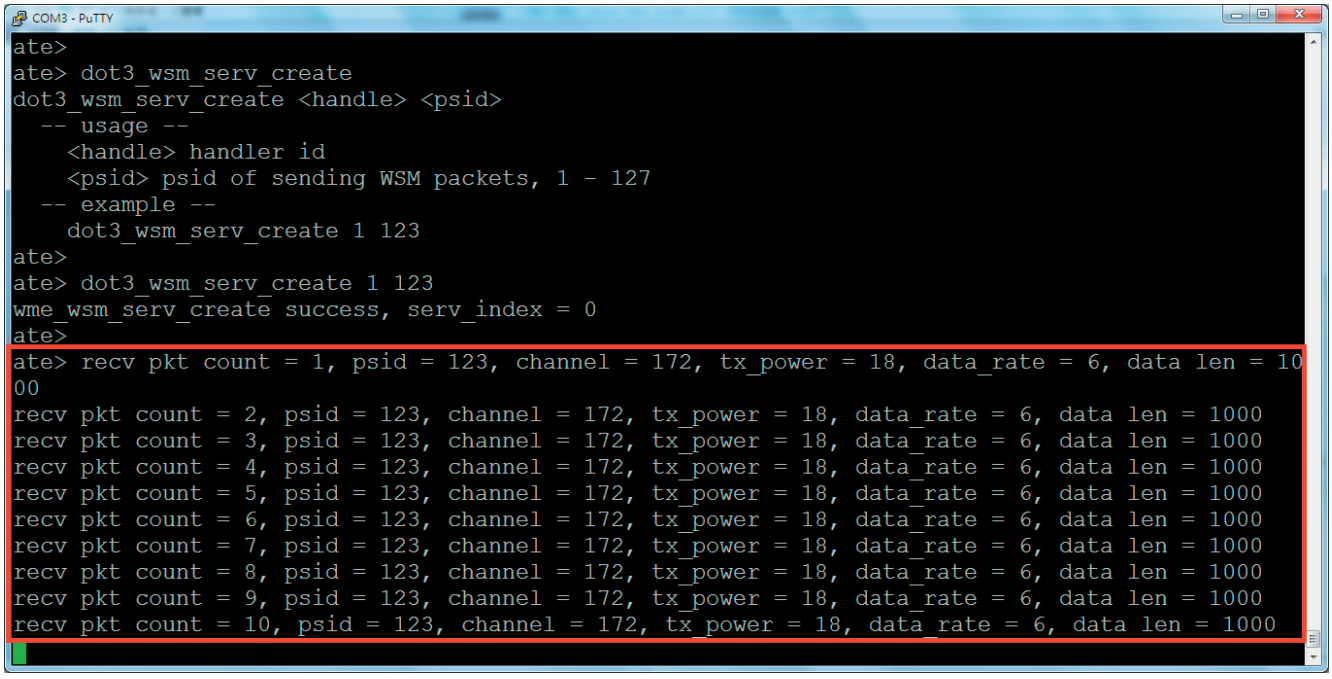


Figure 7 Snapshot of WSM Packets Received

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7. WSM (Indicated by WSA) - Tx & Rx Demonstration

This demonstration will create channel service on channel 172, create provider/user service on this channel to transmit/receive WSA, then transmit/receive WSM on channel indicated in the WSA. It requires configuring two OBU-201 units. Both radio number 1 and 2 are used. Figure 8 illustrates the testing overview. For detail CLI command description, refer to “Application Programming Interface Guide for WAVE Device”.

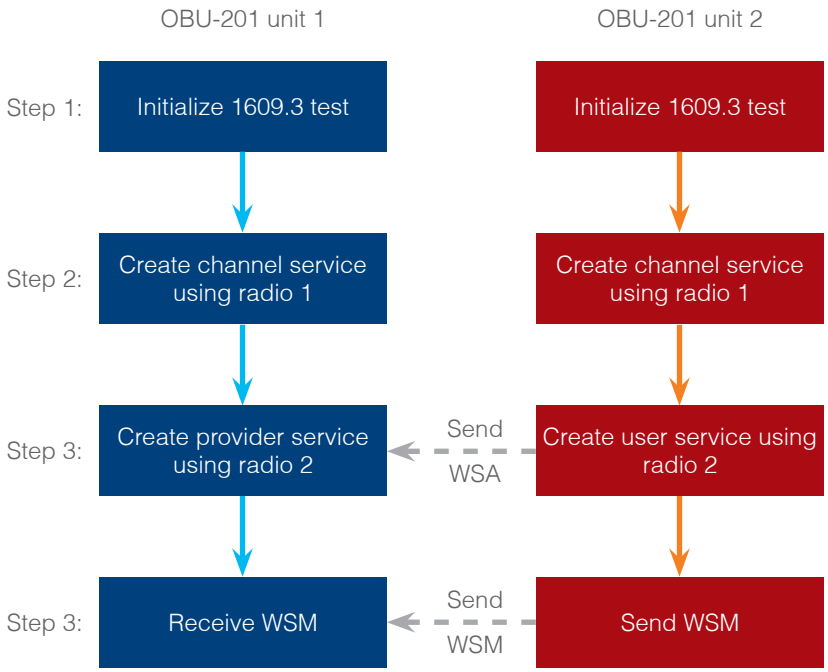


Figure 8 - Channel, Provider, User, and WSM Services

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Step 1:

On OBU-201 unit 1, issue following command to initialize 1609.3 test environment

```
dot3_test_init
```

This command returns a handle id from 1609.3 protocol stack to register other services such as: Provider, User, Channel, WSM

Figure 9 displays the snapshot

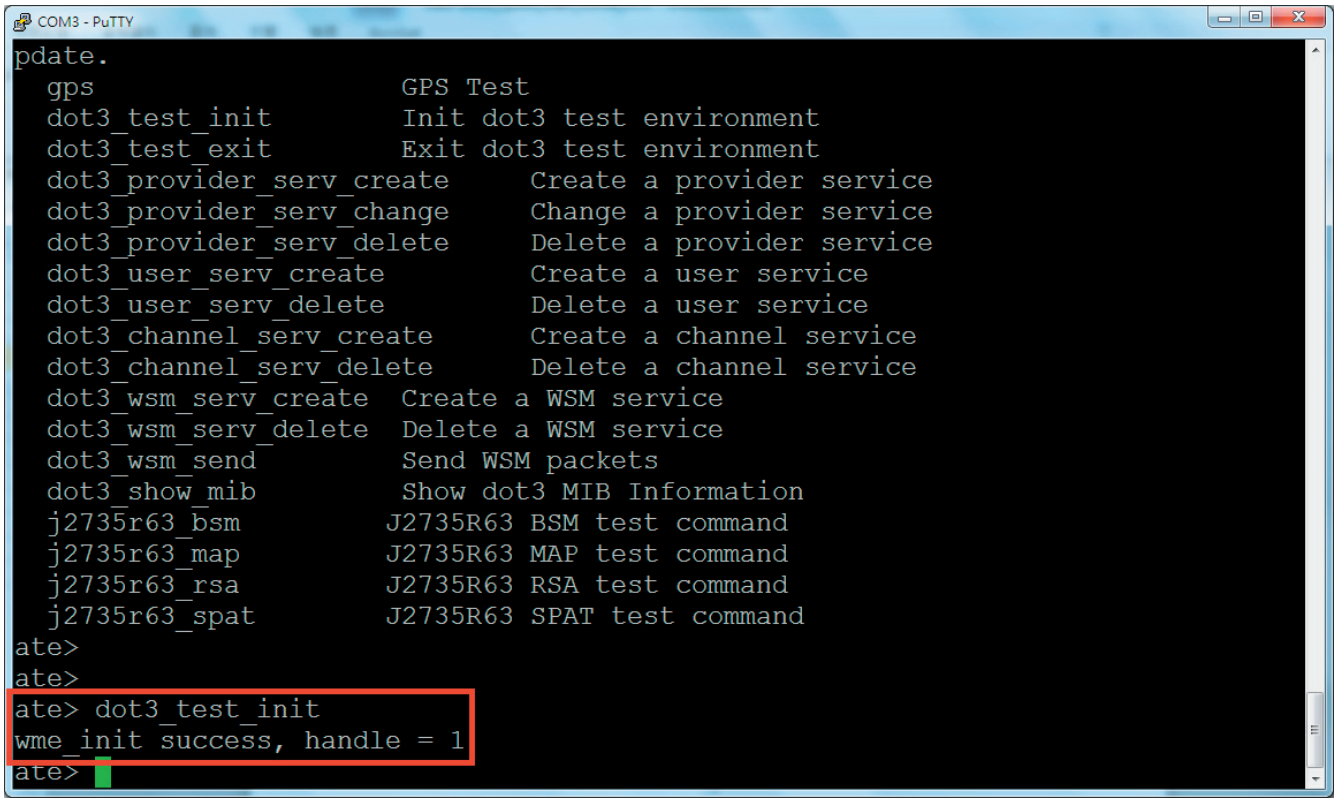


Figure 9 - dot3_test_init Snapshot

On OBU-201 unit 2, apply same command to initialize 1609.3 test environment

Step 2:

On OBU-201 unit 1, issue following command to create a channel service

```
dot3_channel_serv_create 1 1 178 3 5
```

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This command applies handle 1 to create a channel service on Channel 178 using radio number 1. On success, a service index will be returned.

Figure 10 displays the snapshot

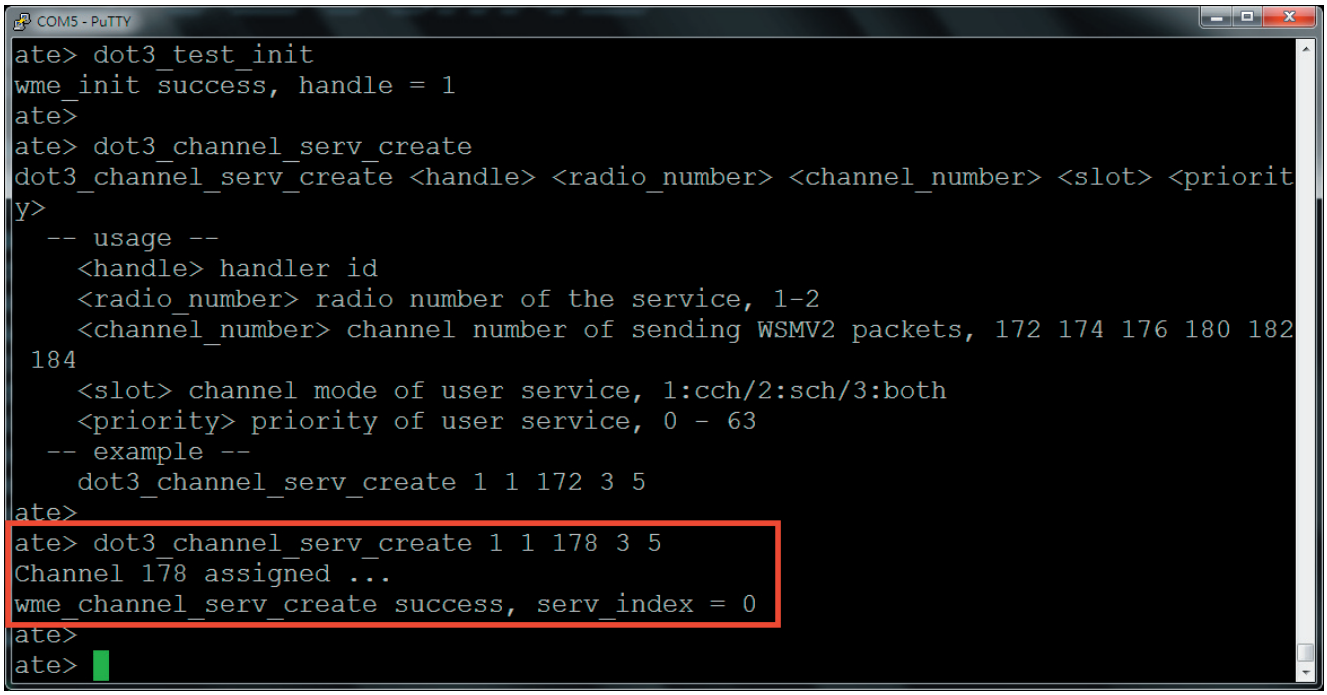


Figure 10 - dot3_channel_serv_create Snapshot

On OBU-201 unit 2, apply same command to create a channel service

Step 3:

On OBU-201 unit 1, issue following command to create a provider service to send WSA

```
dot3_provider_serv_create 1 123 1 2 172 0 5 0 10
```

This command will send WSA on channel 178 and configure channel 172 to receive WSM using radio number 2.

Figure 11 displays the snapshot

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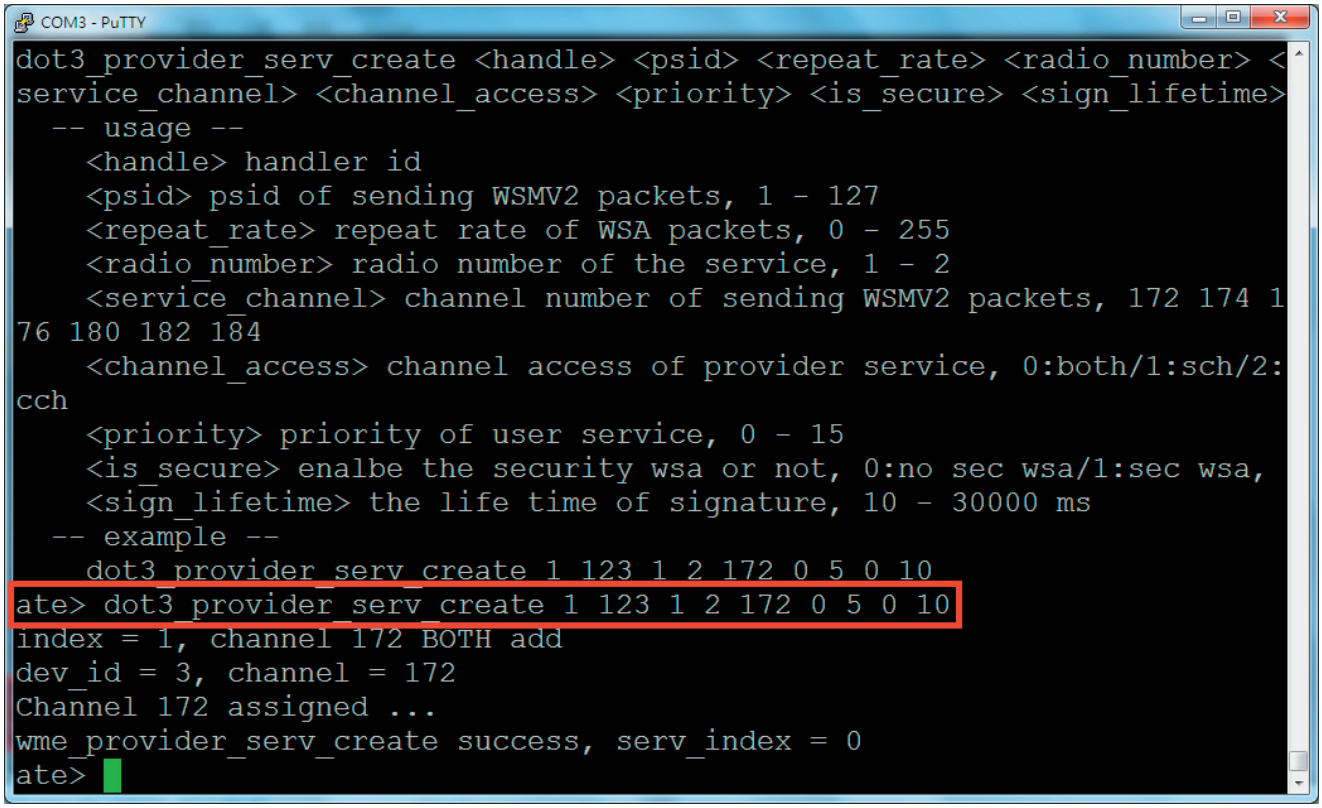


Figure 11 - dot3_provider_serv_create Snapshot

Note:

After testing, we can issue following command (apply handle and service index) to delete provider service

```
dot3_provider_serv_delete 1 0
```

On OBU-201 unit 2, issue following command to create a user service to receive WSA

```
dot3_user_serv_create 1 123 0 5 2 172
```

This command will receive WSA (if PSID & channel number match) and configure channel 172 to send WSM using radio number 2.

Figure 12 displays the snapshot

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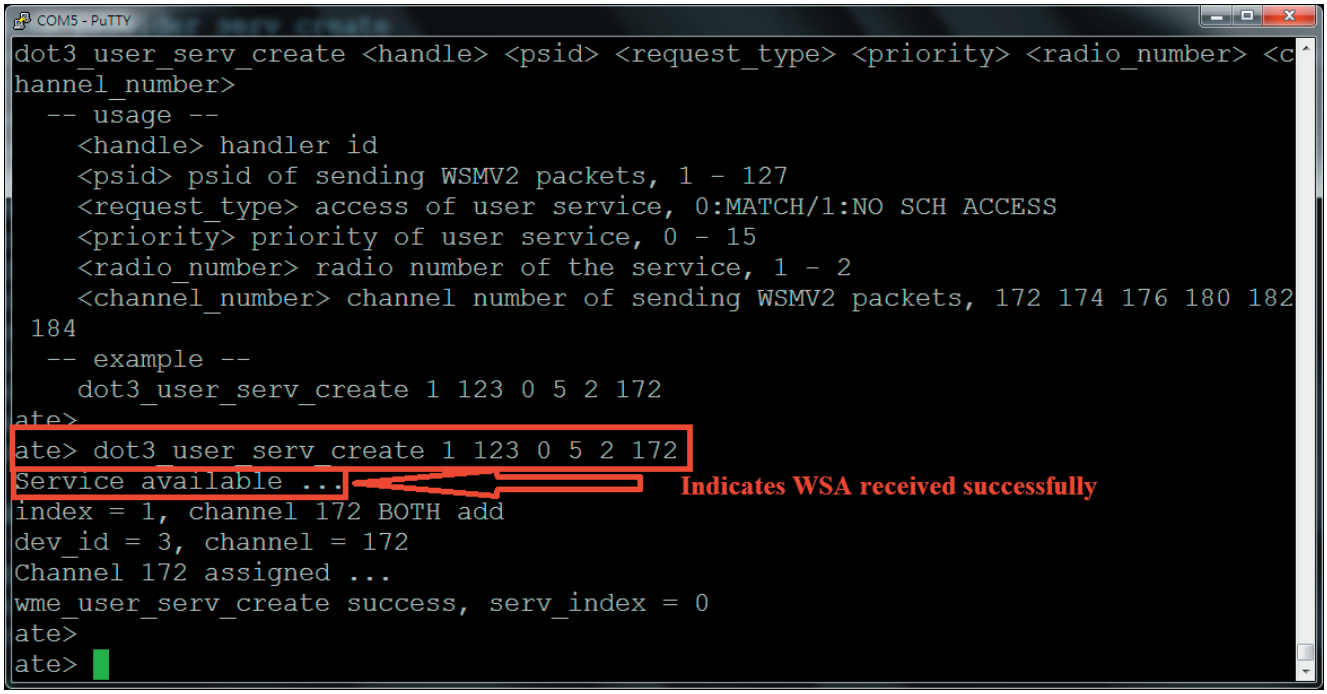


Figure 12 - dot3_user_serv_create Snapshot

Note:

After testing, we can issue following command (apply handle and service index) to delete user service

```
dot3_user_serv_delete 1 0
```

Step 4:

On OBU-201 unit 1, issue following command to create a WSM service

```
dot3_wsm_serv_create 1 123
```

This command will receive all WSM packets with PSID 123

Figure 13 displays the snapshot

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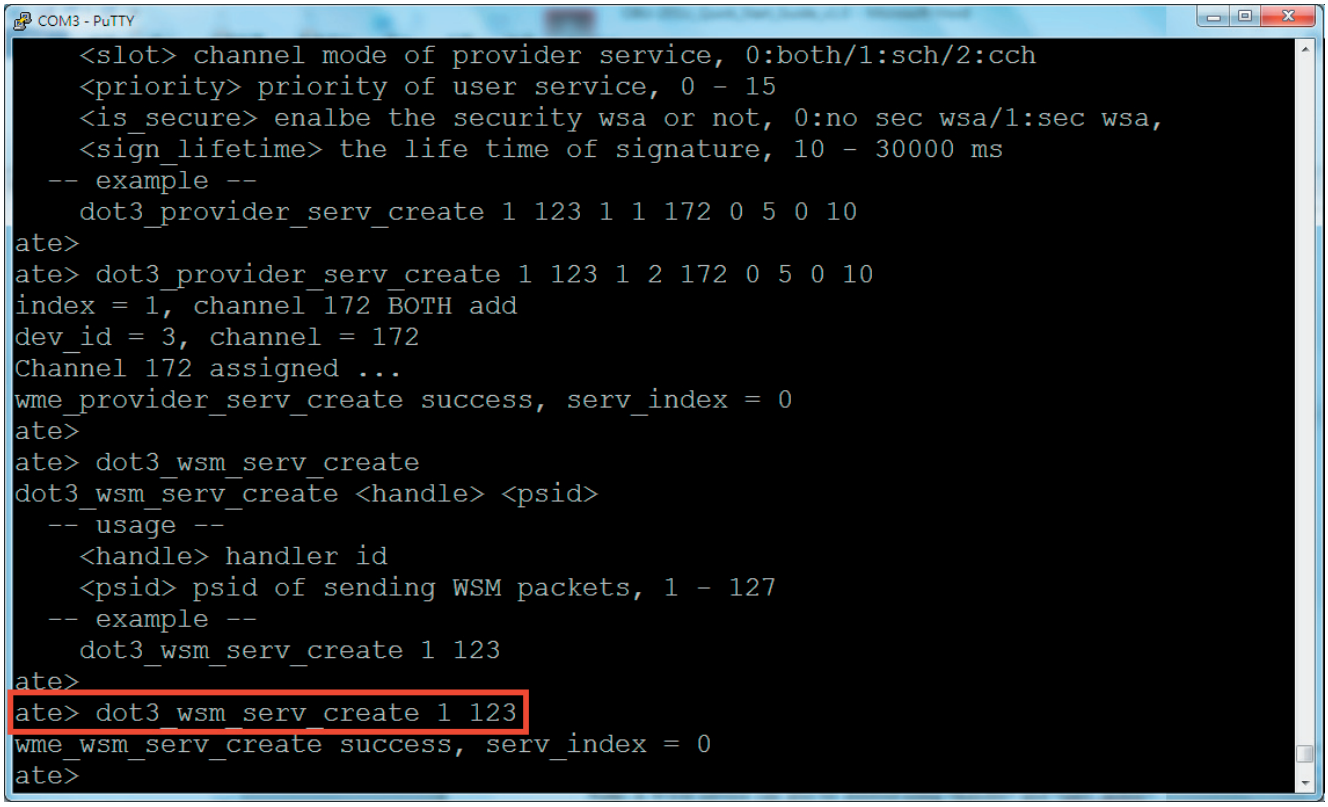


Figure 13 - dot3_wsm_serv_create Snapshot

On OBU-201 unit 2, issue following command to send WSM packets using radio number 2

```
dot3_wsm_send 1 123 2 172 6 18 FF:FF:FF:FF:FF:FF 1000 100 10
```

This command sends 10 WSM packets with PSID 123 on Channel 172 using radio number 2. Figure 14 displays the snapshot

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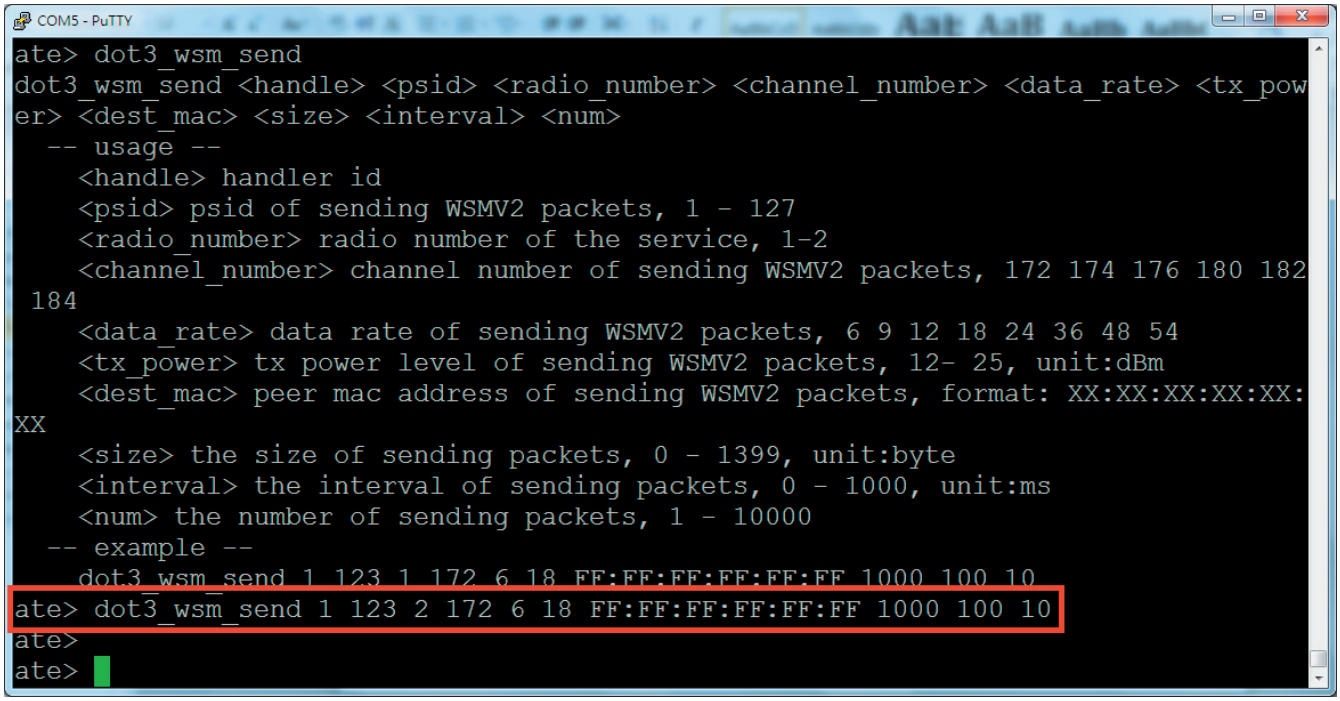


Figure 14 - dot3_wsm_send Snapshot

After execution of above steps, 10 WSM packets are successfully received on OBU-201 unit 1, as shown in Figure 15 (`recv pkt count = 10`):

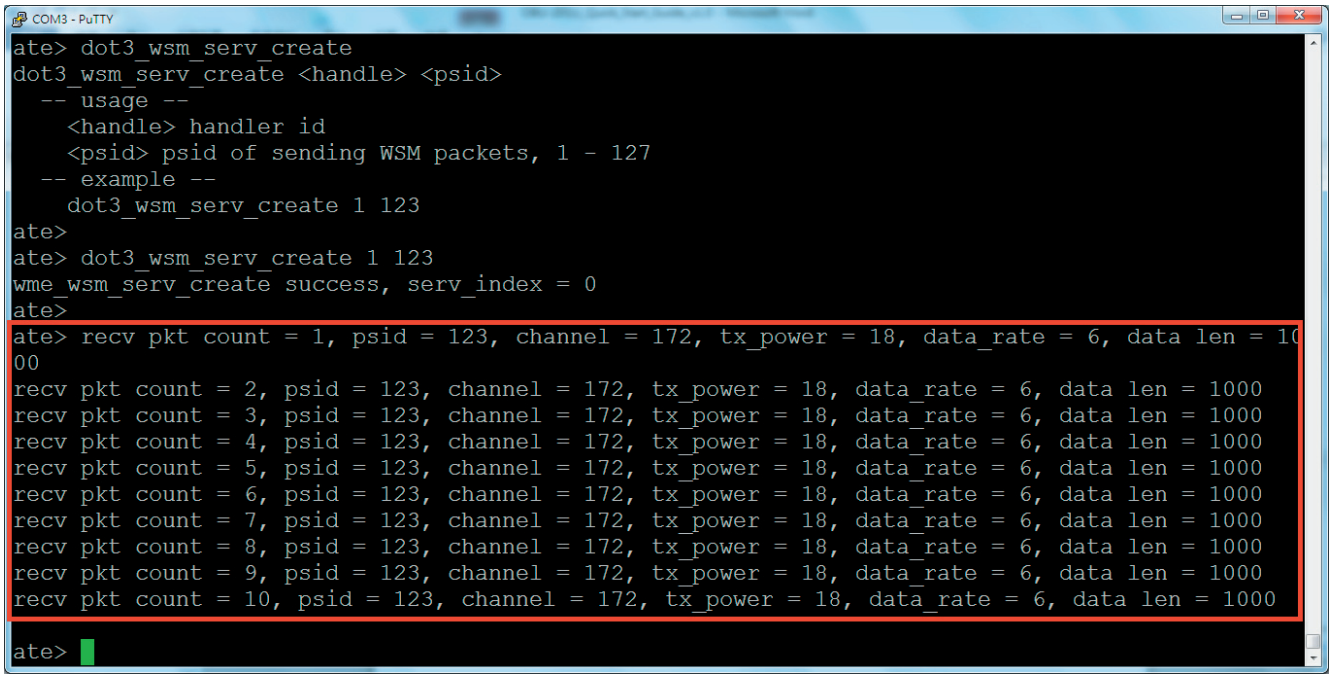


Figure 15 - Snapshot of WAM Packets Received

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8. Checking MIB status

This demonstration illustrates how to retrieve the dot3 MIB information form WAVE device. A primitive "`dot3_show_mib`" is provided to achieve this purpose. This command can be issued at any time to retrieve current MIB information of target device.

Step 1:

Initialize 1609.3 test by issue following:

```
ate> dot3_test_init
wme_init success, handle = 1
```

Step 2:

Create a channel service:

```
ate> dot3_channel_serv_create 1 1 172 3 5
index = 0, channel 178 BOTH delete
dev_id = 0, channel = 170
index = 0, channel 172 BOTH add
dev_id = 0, channel = 172
Channel 172 assigned ...
wme_channel_serv_create success, serv_index = 0
```

Step 3:

Display MIB status:

```
ate> dot3_show_mib
wme_init success, handle = 2
*** Provider entry ***
*** User entry ***
*** Channel entry ***
```

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```
index = 0
priority  = 5
channel  = 172
slot  = 3

*** Wsm entry ***
*** Available entry ***
```

As shown above, we can see a *Channel entry* in MIB.

Figure 16 displays the snapshot

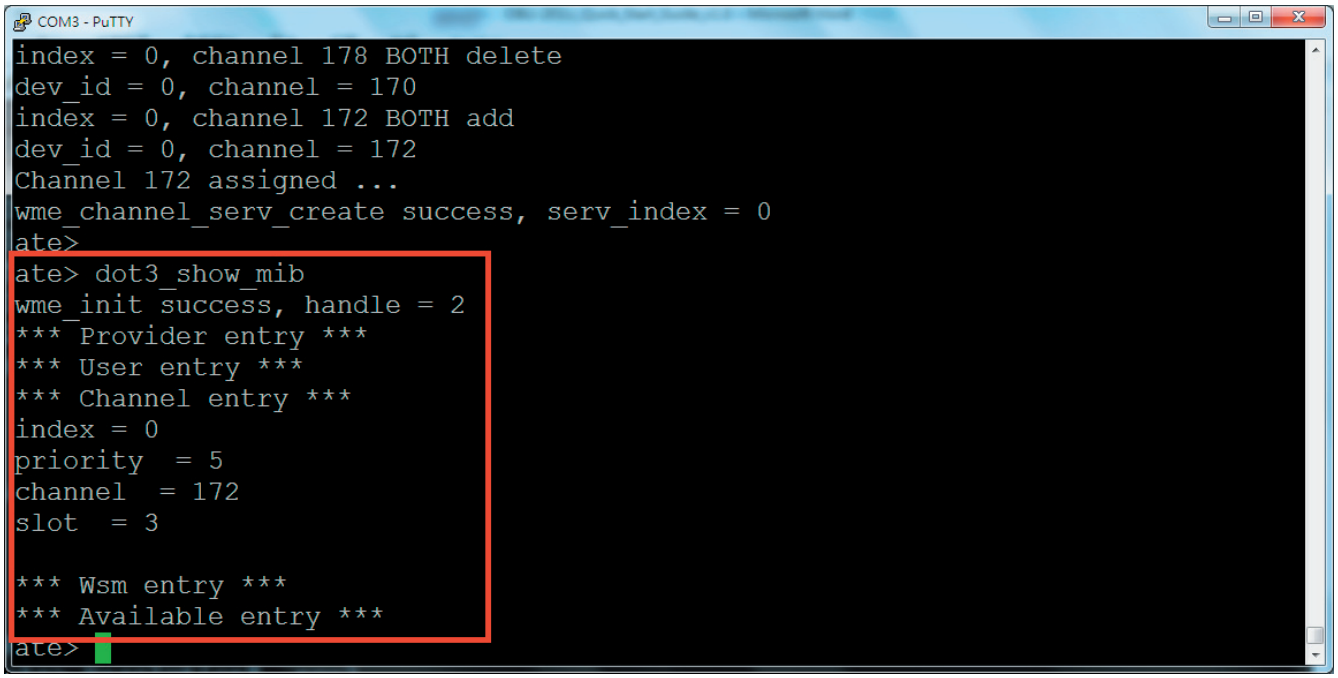


Figure 16 - dot3_show_mib Snapshot

Note:

On receiving WSA, relevant information will be displayed in *Available entry*

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9. J2735 Message - Tx & Rx Demonstration

This chapter demonstrates how to transmit/receive J2735 BSM message. All other J2735 message (SPAT/MAP/TIM/RSA/RTCM/CSR/NMEA/ICA/EVA/PDM/SSM/SRM/PVD/PSM) can also apply the same way to replace the acronyms. For detail CLI command description, refer to “Application Programming Interface Guide for WAVE Device”.

Step 1:

On OBU-201 unit 1, issue following command receive BSM packets

```
rxBsm 123
```

This command will start to receive BSM packets with matching PSID (for 5 seconds)

Figure 17 displays the snapshot

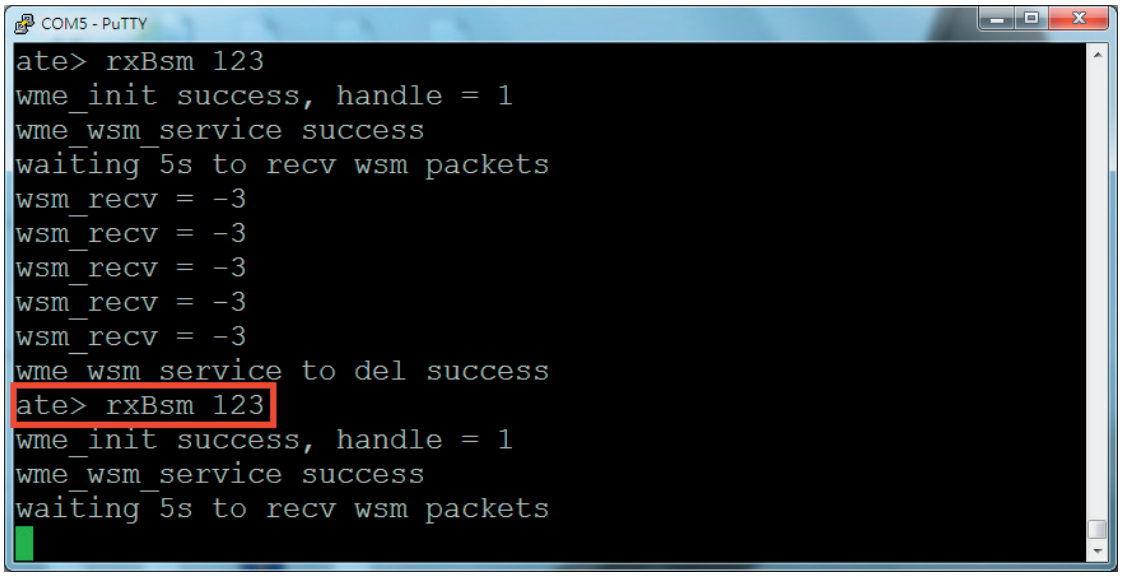


Figure 17 - rxBsm Snapshot

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Step 2:

On OBU-201 unit 1, issue following command to send BSM packets

```
txBsm 123 6 18 FF:FF:FF:FF:FF:FF 1000 1
```

This command will send 1 BSM packet with PSID 123

Figure 18 displays the snapshot

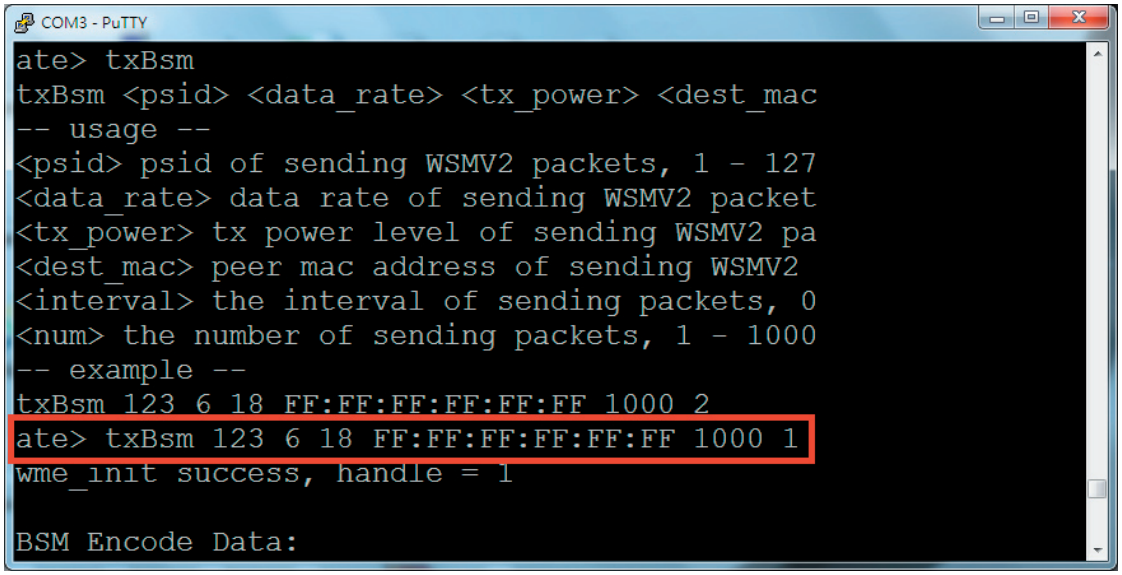


Figure 18 - txBsm Snapshot

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After execution above, we have successfully received BSM packet, as show in Figure 19

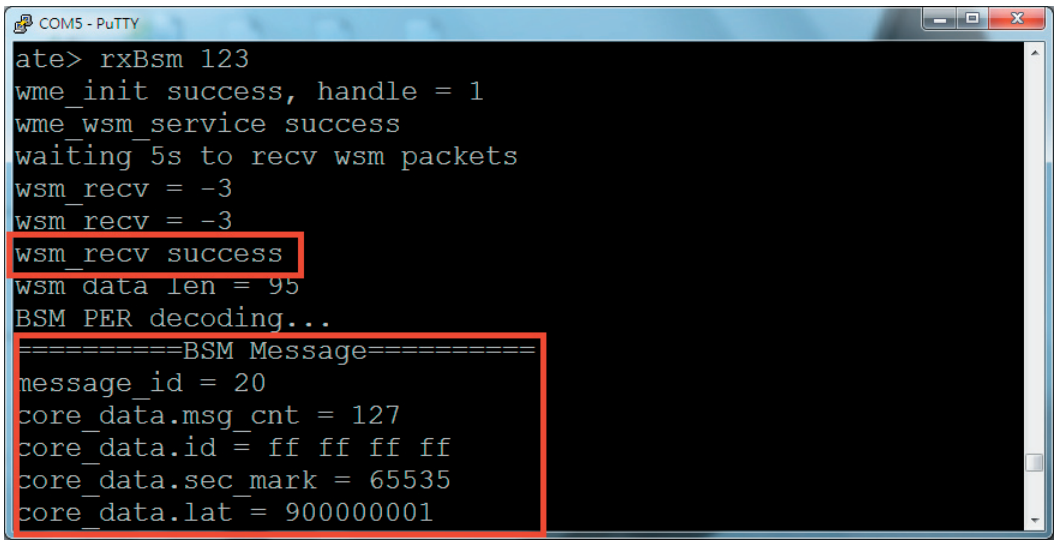


Figure 19 - Snapshot of BSM Packet received

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10. Firmware Upgrade

For firmware upgrade, an additional ethernet cable need to be connected between PC and WAVE device. Or connect both ethernet ports to a common switch/hub. The IP set in PC and WAVE device should be in the same sub-net. In this chapter, we assume the IP address of WAVE device set to "192.168.1.30", and the IP address of PC set to "192.168.1.10".

Two methods of firmware flashing are supported on Unex WAVE devices:

» CLI command - enables flashing of firmware image located in RAM partition. A system reboot is required for the new firmware to load.

» U-Boot - requires knowledge of the address and size of the main firmware image partition

For ease of use, we recommend applying CLI command to flash the firmware.

10.1. Using CLI Command

Steps:

1. Setup a serial connection between PC and WAVE device. Turn on the WAVE device, do not hit any key until the prompt "ate>" appears in the terminal.
2. Ensure the ethernet connection. Issue following command to get Ethernet IP address of device

   ```
   ate> eth ip
   Ethernet IP address is 192.168.1.30
   ```

   Configure IP address of PC as "192.168.1.10"

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3. Access the ftp server running on WAVE device (apply FTP client application, or type "`ftp://192.168.1.30/`" in explorer), upload the firmware binary (assuming named as "`wave-sdk.img`") to **disk D**: (RAM partition) of WAVE device.

4. Issue following command to flash the firmware:

   ```
   ate> firmware update D:/wave-sdk.img
   ```

   Note: This step may consume about 40 seconds.

5. Reboot device.

   ```
   ate> reboot
   ```

10.2. Using U-Boot

Steps:

1. Setup a serial connection between PC and WAVE device. Reset the WAVE device, or issue "`ate> reboot`" in ate mode. Hit any key when following appear within 3 seconds.

   ```
   Hit any key to stop autoboot:  3
   ```

   When following prompt appears, indicated entering U-Boot mode

   ```
   U-Boot>
   ```

2. Ensure the ethernet connection. Issue following command to check parameters of Ethernet IP address of device and server IP address (PC running TFTP server)

   ```
   U-Boot> printenv
   ...
   ipaddr=192.168.1.30
   ...
   serverip=192.168.1.10
   ...
   ```

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3. Configure IP address of PC to "serverip". Verify the connection by issuing "ping" command in U-Boot mode

```
U-Boot> ping 192.168.1.10
Link: UP
Duplex: FULL
Speed 100BASE-X
Using  device
host 192.168.1.10 is alive
```

4. Running a TFTP server application (ex. Tftpd64) on PC. Issue following command in U-Boot mode to download binary image (assuming named as "wave-sdk.img") from PC to device.

```
tftp wave-sdk.img
```

5. Issue following to erase flash memory

```
mw.l 0x41016850 0x001f0000
protect off 80000 1bffffff
erase 80000 1bffffff
```

6. Issue following to replace existing firmware image wth the new one:

```
cp.b ${fileaddr} 80000 ${filesize}
setenv bootcmd 'cp.b 80000 ${fileaddr} ${filesize}; bootm'
saveenv
```

7. Finally, reset the device by issuing:

```
reset
```

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